

Excerpt, Chapter 4
On Summoning Demon Gods: AI in the
Economy

**TAYLORISM CRUSHES IT: GENERATIVE AI AND
DESKILLING THE LONG TAIL**

In the summer of 2024, Apple aired a new video ad for its latest tablet iteration.¹⁵ The ad begins by showing us a collection of tools, whose use clearly requires skill and knowledge, arranged into a sleek pyramid. Most of the pieces are art-related. A trumpet crowns the arrangement. A metronome ticks implacably and ominously, perhaps as if saying “Your time is running out.” There are other instruments, like a piano and a drum set, alongside studio monitors, and a mixing table. Tools of the visual arts are equally prominently displayed: various paints, a poseable artist’s mannequin and easels, a film clapper and cameras. An architectural diorama, and notebooks also have their place, as does a blackboard with some nonsensical (?) math. Alas, the arrangement sits atop the platform of a hydraulic press, and gets slowly, painstakingly crushed. Paint gushes, piano keys shear and splinter, glass shards of camera lenses spray out explosively, the trumpet gets bent and contorts into an unrecognizable knot. When the press lifts up again, all that remains is a tablet. All the tools, all the skills, art and love they represent, have been symbolically—violently—transferred into the iPad.

Apple's ad triggered a wave of backlash. It had been all too real. The aim of transferring knowledge and skills into tools was laid perhaps a little too bare. The ad, called "Crush!" (even the name said the silent thing out loud), inadvertently gave the game away: in transferring all the skills represented by all those tools and materials into the tablet, human agency and the human meaning derived from honing and using those skills, got bulldozed, devalued, and disappeared. What used to be not just any skills, but skills deeply representative of the possibility of taking pride and pleasure in one's work, was flattened into yet another commodity—"an absolute powerhouse for AI" available on apple.com, starting at \$999.

The advertisers may have been oblivious to it, but to plenty of people one thing was obvious: the Apple ad represented a devaluing of knowledge and skills. Taylorism strives to centralize knowledge in a managerial bureaucracy, in order to separate out tasks, and deskill them. The Apple ad symbolizes a new and blatant form of techno-Taylorist deskilling—the transfer of knowledge and skills not into a managerial bureaucracy, but directly into a tool, into a machine. In this case, the iPad took center stage—but everyone understood that this ad was about that one flashy technology that is most emblematic of artificial intelligence today: generative AI.

Think about generative AIs for art and design that generate images from text input, like Dall-E or Midjourney (trained, as so often, on the private property of millions of artists, and on global as well as local commons). Think about AIs for writing. Think about AIs that are supposed to find and summarize legal cases. Think about AIs aimed at writing computer code, like Microsoft's Copilot. Think about education AIs that are supposed to aid or replace tutors. The list of domains that generative AIs are being developed for can seem endless. Any task whose aim is the production of digital objects—from text to audio, from images to movement instructions for a robotic arm, is apparently up for grabs by generative AI.

One can and should harbor doubts about how successful models based on the current crop of machine learning techniques will ultimately be—but there is little reason to doubt what the

aim is: to embed knowledge in machines—knowledge that previously belonged to the people working in these domains. The nature of the human knowledge that is targeted varies greatly, within and across domains. It might be abstract or concrete, it may be declarative or procedural. But regardless of its nature in these or other regards—from an economic perspective, it is usually knowledge that counts as potentially marketable skills, knowledge that is constitutive of the primary asset that the workers in these domains have to sell: their ability to perform particular work. In other words, just as Amazon’s warehouse AI embedded a surveillance drive of management in a machine, so generative AI embeds knowledge of tasks (and the drive to centralize it) in a machine. Both the surveillance AI and the various generative AI approaches realize the same drive towards lower wages that already brought the steam engine into the cities—the need to make workers replaceable by others, to lower wages by increasing labor supply. The method of generative AI is technological Taylorism. The point is—to deskill.

MAKING GOOD JOBS BAD

What kinds of effects will this deskilled, “AI-powered” work have? The question is probably worth its own book, but one thing that seems obvious, is that it will make things cheaper—both in terms of price, and in terms of quality—and that this will further exacerbate pre-existing inequalities in many of the domains targeted by generative AI. Most work that can be deskilled by generative AI will be increasingly gigified, while a small section of specialists will produce AI-free and “artisanally” for the upper crust. So-called self-driving cars already operate on a mix of automation and gig-work. The “autonomous” vehicles of General Motors subsidiary Cruise, for instance, are remotely assisted some 2–4 percent of the time, or every 2.5–5 miles, with their human “autonomous vehicle operators” continually monitoring many cars simultaneously.¹⁶ The rich, of course, have their driving produced artisanally already—by chauffeurs. AI tutors will be running on language models, but just as with Amazon’s warehouse surveillance and autonomous cars, AI tutoring will also be supported by a hidden underclass of deskilled

gig-workers that monitor the chat windows of dozens of students at the same time. Meanwhile, the wealthy will continue to hire Ivy League PhD students to look after their children.

The legal domain, just like education, is already starkly divided for the poor and the rich: a wealthy defendant may well have a whole team of people working full time on their case, while poor defendants will be lumped in with a dozen others on a public defender's docket. AI legal assistants and AI legal search engines will likely lower the wages for paralegals (who are being deskilled). "Ambulance chasers" and "sharks" will increase their dominance of the low-end market, cutting costs due to AI, while also undermining quality. Competitive pressures mean that honest and dishonest lawyers alike will have to take on the absolute maximum number of cases that can be done in a certain amount of time. Those lawyers who genuinely care about serving the underserved will suffer from the increasing pressures of this competition in their market: AI-assisted competition will drive up the numbers of cases these lawyers have to take on, and drive down the quality that they can afford to provide. Meanwhile, wealthy clients and corporations will be largely unaffected, since they will continue to be served by law firms whose services are way out of reach for almost everyone else.

In the realm of coding, big tech companies will continue to pay high wages to those who produce code for high stakes situations, where even small optimizations can be worth a lot of money. But an increasing number of programmers will be subjected to competition from companies that provide minimal training (say, a four-week course in using generative AI for programming), and that treat their workers as replaceable and disposable, gigifying and deskilling ever more coding work.

In sum, in many of the domains where generative AI may play a major role, we think the effects of AI automation and deskilling will likely be an acceleration, intensification, and spreading of tendencies that we can already observe today, such as gigification: it will make low-quality work somewhat worse, but much cheaper, thus creating downward pressure on wages and quality alike, while putting high-quality work increasingly out of reach. Out of reach, both for all those workers who would much rather do such work,

and for those consumers who cannot afford to buy anything but things and services produced in AI-assisted contexts.

Generative AI will bring this kind of crowding out of the mid-level quality products, and the (often good) jobs associated with them to new parts of the economy. At the same time, the pattern is a more general one that has been quite typical of capitalist development for decades now, and another aspect to the deskilling drive. From shoes to tools to pots and pans—the medium-quality market has been crowded out by a lower-quality level that is often much cheaper (and most of the time simultaneously less durable), while the higher level quality goods are increasingly out of reach for most people. Perhaps the most obvious example is furniture—ready-to-assemble furniture from IKEA and its copycats are ubiquitous, but furniture that could last for more than one move is increasingly limited to a high-end market. The high-end market may be largely unaffected, but the mid-level quality often cannot compete with the lower end. This is especially true in a country like the US where the majority of people live paycheck-to-paycheck and find saving almost impossible. Often, cheaper goods have to be replaced significantly more frequently, perhaps even to the point where the cheaper goods end up costing more money in the long run. It's expensive to be poor.

Clearly, all of this will further stratify the working class. It will create more separation between “good” and “bad” jobs and increasingly differentiate pay levels. It will make the experience of being a worker under capitalism ever more different according to which stratum of the working class one is in. The stratification of tomorrow grows directly out of the stratification of today, and deskilling is, in fact, a major reason for working class stratification as it exists now. Already, in many professions, we find a spectrum from highly deskilled work aimed at producing lower cost goods, to workers with deep knowledge of the whole labor process, who produce high-end, or luxury commodities and who have higher incomes. Consider, say, a watchmaker. While most of us use watches mass produced in industrial settings (or simply our phones), prices for “artisan-made” luxury watches can easily outpace the lifetime earnings of most workers. Differentiation-by-deskilling.

What the spectrum from deskilled to “artisanal” work looks like, both from the producer’s and the consumer’s perspective, is of course quite specific to the particular object. We could contrast watches, for instance, with restaurant meals. In restaurants, deskilling is often achieved simply through the organization of work, by management. That is, tasks are split up, and assigned to people of different wage levels—say, bussers, prep cooks, line cooks, sous chefs etc. Again, it’s important to stress here that “deskilled” is a technical term that has little to do with whether a worker has skill, and far more to do with how easy it is to replace them in a particular position: any prep cook is highly trained and skilled in a number of domains, and most can easily out-chop anyone without that training; in fact, these skills are a job requirement, whether the position is deskilled through organization/management or not. But the way that the labor process is divided is aimed at deskilling the prep cook, and concentrating knowledge of the whole of the labor process with management. In the case of a restaurant, management might simply be the chef manager, who sets the restaurant’s menu, is relevant to the restaurant’s brand, etc. Of course, the chef manager may well be the actual capitalist of their own small capital, if we are talking about a single restaurant. In the fast food industry, i.e., in the most gified part of the restaurant industry with the most mass-produced (and often cheapest) food, organizational deskilling is accompanied by technological deskilling. Burger patties and french fries, for instance, may be prepared in fully industrial settings, and arrive at the restaurants in frozen form. Since this allows for intense standardization and maximum task splitting, the work performed by those restaurant workers, too, is maximally deskilled—that is, their performance and their movements can be tracked according to detailed standards, and management can often treat people not as humans, but as replaceable inputs in their production process.

Even in the general public discourse, the derogatory “flipping burgers” is often used to denote bad jobs that have been subjected to deskilling to a very high degree—but of course that was never in the hands of the individuals who work there. Deskilling, continually chased after by capital, produces a simultaneous need for a labor pool deprived of other opportunities. Fast food workers,

of course, have correctly identified these facts, and are joining forces. They have been courageously fighting back, for instance through forming the Fight for \$15 movement that demands a right to a union and a higher minimum wage.

The transfer of knowledge into tools that is involved in deskilling also produces a second source of working class stratification: whenever knowledge is embedded into tools, an associated kind of specialist emerges, people who do have access to the relevant knowledge, who can further develop the tools, who can service and customize them. In the case of the steam engine, this included, for example, the machinists who could make the relevant parts. Of course, just as with the pyramid of managers, the expanding specialist labor may subsequently itself be subject to deskilling, as happened with machinists. But broadly speaking, deskilling tends to be accompanied by the creation of its own techno-bureaucracy, responsible both for the tools, and for concentrating knowledge relevant to them. Sometimes, the members of this new bureaucracy are first drawn from the workers themselves, sometimes they come from outside.* Regardless, they are responsible for putting into the service of capital both the knowledge embedded in the tools, and the managerial knowledge of the tasks (and their splitting up) that the tools require. In other words, they will form a specialized group whose job is in an ambiguous field between engineering and management, as evident for example in the many degrees in fields like “engineering management,” “business engineering,” “technology management,” “systems engineering” (a term that can, perhaps not incidentally, be traced back

* Indeed, Frederick Winslow Taylor himself became a manager, and then the founder of scientific management on such a path. While from a wealthy family (his father was a Princeton-educated lawyer), Taylor decided to forgo his Harvard admission, and become a machinist instead. After an apprenticeship, he started out as a workman at Midvale Steel Works. Then, however, he became the gang boss of his former coworkers, who soon called him a tyrant. From the moment he took on that position, he continued to develop his basic ideas: that any control that workers have over the processes of work stand in the way of productivity, of capital pressing the most labor out of any person hired, and that hence, all such knowledge must be concentrated in management.

to Bell Labs, the birthplace of the transistor), and many more like them. Any company will happily pay higher wages for these “upskilled” workers, as long as it makes up for the lowering in wages, and the potential increase in speed, control, productivity, and so on elsewhere. Perhaps a company could either pay ten workers 50k each, without deskilling them, or it could pay one worker (turned techno-bureaucrat) double, and nine deskilled workers half. This is the real life version of the industry fable of the so-called “10x engineer,” a mythical engineer that allegedly adds 10 times the value of a normal one to a company, and the real mechanism behind it: the value that is “added” is literally the wages that the other nine workers are no longer paid, and which thus remain on the credit side of the company’s ledger. Whenever new technology enables such an equation to be drawn up, capital will happily produce highly paid positions—provided, it saves enough money elsewhere.

As we live in times where the promise of social mobility through education is increasingly turning stale, with many people, particularly in the US, buried under suffocating piles of student loans, perhaps it’s worth reconsidering whether that promise wasn’t always based on the fact that specialists are often the flipside of a deskilling process for others. That is, perhaps it was always impossible for that path to be plausibly available for everyone—after all, most people have to be on the deskilled side for the equation to work out in capital’s favor.

MAKING TOOLMAKING TOOLS

What does transfer of knowledge look like on the side of the machines? At a very abstract level, this is simply the transition from special purpose to (more) general purpose machines, or the ever increasing ability of machines to be programmed, and to process more and more information. And indeed, that would be an excellent way of re-telling the story of the transistor. Although written as early as 1974, we can draw, again, on Harry Braverman’s book—perhaps sometimes historical processes are easier to understand early on, than when they have developed all kinds of complexities. In discussing automation in the metal industry,

Braverman (himself a former metal worker) suggests that automation first targets objects that are produced on a very large scale: it tends to be worth investing into machines for making things like screws or standard car parts. But those screw machines themselves contain many specialized parts, each made only a few times. Since we need far fewer screw machines than screws, screw machine parts in turn will be made in much smaller batches (some perhaps will even be unique). As it turns out, far more objects are produced in small batches (like screw machine parts), than in large batches (like screws). And because they are produced only in smaller quantities, it is harder to automate their production cost-effectively.

Imagine a graph of all objects produced in a year, ordered by how many times they are produced. Picture the graph with count of individual objects produced on the Y-axis, and the individual objects on the X-axis. Let's say the objects are listed in increasing order of production count. Our graph would start with a very long list of objects that were produced only once, then a long list of objects that were produced twice, etc., until it ends with the object produced most often. A distribution like that is said to have a *long tail* of many, many objects produced in small numbers. Automation will tend to target the right side of the graph first, aimed at those objects that are produced in huge numbers, and it will involve special purpose machines, like a screw machine, designed for the production of that one object, in massive quantities.

What about the long tail, all the objects that are produced only once or a few times? This is often a place for knowledgeable specialists, people who have the skills to use a variety of tools to produce highly specific results. In the case of our screw machine parts, it might have been machinists who knew how to precisely cut metal into the complex shapes needed. But the necessity of that knowledge increased the machinists' bargaining power, and hence tended to produce higher wages for them. We know, by now, what shape capital's solution to such a "problem" takes: transfer knowledge to the machine! In this case, such a transfer of knowledge was originally known as *numerical control*, i.e., machines that could process a certain amount of numerical information.

On the one hand, machines were equipped with ever more monitoring devices and sensors that could provide information about the current state of their task. On the other hand, machines were increasingly programmable, originally with punch cards—the same kind of technology that allows for player pianos to be programmed to play different songs (as in Vonnegut’s novel)—then increasingly with transistor-based chips. Numerical control machines turned into computer numerical control machines, or CNC machines, which could cut increasingly complex shapes. They could take over more and more calculations (say, the path along which a cutting tool must move to create a particular curvature, radius, etc.) that had previously been part of the machinist’s job. Because they were programmable and increasingly information processing, they could fulfill an increasingly larger set of functions, and hence make an increasing variety of objects. Today, even YouTube is full of videos of CNC machines carving wood or metal blocks into fascinating and unique objects, from screws that turn both ways to spectacular marble runs. Unlike our screw machine, then, these machines can make a great variety of pieces. Or, to put it another way: as we go down the long tail, we move from objects created in large numbers by special purpose machines to objects created in small numbers by (more) general purpose machines.

Automation by AI is very much a continuation of this trend. Generative AI is aimed at automating the production of many things at the very end of the tail, from newspaper articles to logo designs—that is, things that are really only produced once. Take even the term “general” in “artificial general intelligence.” Clearly, if AGI has a non-occult meaning (i.e., if it still denotes a tool rather than a myth) the “general” in it is the one from “general purpose.” The development of such a fully general purpose machine might be considered the holy grail of AI research and development. Even “ChatGPT” might be a pun on this: while GPT stands here for *generative pretrained transformer*, a language model architecture, in economics GPT is often short for *general purpose technology*, technologies with extremely wide applications like the steam engine, electricity, or the transistor. And while that may or may not have been a sneaky reference to that envisioned

general purpose character, sometimes the thing is said quite explicitly. Take the definition from the charter of OpenAI: “artificial general intelligence (AGI)—by which we mean highly autonomous systems that outperform humans at most economically valuable work.”¹⁷ Clearly, they are talking about a general purpose machine, one that could (if it existed) replace *homo sapiens* the toolmaker with itself, the toolmaking tool.

MACHINES DESTINED TO BE DISPOSED OF

A world dominated by such hypothetical tools would seem to have little room for humans to act purposefully on the world, or derive any meaning from life at all. As OpenAI CEO Sam Altman once said: “AI will probably most likely lead to the end of the world, but in the meantime, there’ll be great companies.”¹⁸ If the aim seems absurd, we must remember that it represents the logical endpoint of one of capitalism’s tendencies, the drive to deskill, to transfer knowledge from workers to machines, from humans to capital. Speculations about AGI are, in this sense, speculations about the endpoint of the drive to deskill: ultimately targeting *all* human knowledge. That is not to say that this is a plausible outcome; there are many countervailing forces (not least the fact that we continually generate new knowledge and new tasks). Indeed, capitalism’s relationship to knowledge is rather paradoxical: on the one hand, there is a tendency to universalize certain kinds of knowledge, to make sure everyone is in possession of it—since that, too, is a form of deskilling. That is, if everyone knows a thing, then no group of workers can gain special bargaining power from possessing that knowledge. We can take reading and writing as an example, and we could look at many lobbying efforts and political arguments over what should be included in school curricula through that lens. Even the investments that many big tech companies have been making into their own kind of schools, so-called coding academies, can be understood in the light of these universalizing pressures: make everyone into a programmer, and programmers can no longer demand higher-than-average wages.

On the other hand, capitalism produces precisely the opposite relation to knowledge, one that is decidedly not that of an

open and democratic society. We might say that capitalism's other relation to knowledge resembles nothing so much as an intelligence agency. That is, it is incessantly involved in gathering and producing knowledge, while also putting all that knowledge on a "need to know" level of security, concentrating it, and sharing it only when its aim (profit) cannot be otherwise achieved. This is the movement that continually transfers knowledge from workers into tools, from people into capital. If the "general" in AGI is the one from "general purpose," then this, surely, is the actual meaning of the "intelligence" in A(G)I: the kind of intelligence you gather, not the kind that you possess.

Braverman described this second aspect, the use of science in ways entirely antithetical to science itself: "The more science is incorporated into the labor process, the less the worker understands of the process; the more sophisticated an intellectual product the machine becomes, the less control and comprehension of the machine the worker has." These two ways of treating knowledge can seem utterly contradictory—only if we conceive of them both as forms of deskilling, of undermining the bargaining power of workers, can we make sense of this apparent paradox.

Let's take another look at that definition of AGI—"highly autonomous systems that outperform humans at most economically valuable work." That "outperform" is a rather telling word, isn't it? Under no organization of society, other than capitalism, are the tools somehow conceived of as being in competition with the tool-users. The philosopher Shannon Vallor has discussed this idea of competition, and the way it impoverishes our sense of self, our idea of meaning, and even what it means to be human: "those invested in the maximally efficient extraction of productive outputs from human bodies have been trying to get us to view ourselves—and more importantly one another—as flawed, inefficient, fungible machines destined to be disposed of as soon as our output rate slips below an expected peak or the moment a more productive machine can be found to step in."¹⁹ She's certainly right. We might even go one step further—it is not simply that those invested in the maximally efficient extraction of productive outputs (those that we call capitalists or the representatives of capital in this book) try to get us to view ourselves this way.

It is also that this reduction of our humanity is inherent to the semi-voluntary mode of “cooperation” that is wage labor; to capital, we are always a means to an end (profit), and never partners in a project that would involve its own meaning (regardless of what HR departments may tell us). In chasing profits, in seeking to maximize its measure of success, capitalism must seek to press the most productivity out of tools and people alike. And what is capital’s drive to deskilling, if not the drive to make us ever more fungible (interchangeable and substitutable)?

Of course, the fungibility that deskilling is aimed at, is about making workers replaceable by other workers, while automation is ultimately aimed at substituting tools for people. In that sense, under capitalism, the competition between worker and tool that the word “outperform” implies is real. When a task is partly automated, when productivity is increased, the people who used to do all that now superfluous work do not generally rejoice that they have won free time to spend on something else. Instead, the people themselves are made superfluous to the enterprise they were formerly engaged in. That is what, in Vallor’s words, turns people into “machines destined to be disposed of.” Some, maybe even all of the replaced people get fired, and the savings in time become the source for extra profits for capitalists, and extra distress for the workers.

One of the truly strange properties of capitalism, isn’t it? That it continually makes people superfluous, rather than merely their work, setting people free not from boring toil, but from their ability to make a living. In a society that would deserve the label “free,” perhaps automation would be aimed not at making people superfluous, but at eliminating the aspects of work that are onerous, that feel like drudgery. And all the scientific and engineering efforts that go into automation could be aimed not at deskilling, not as something set against the workers, but aimed at increasing the creative, expressive, social, scientific, and otherwise fulfilling parts of work. It could increase human agency over the labor process, rather than concentrating it within managerial structures and machines. Alas, capitalism is no such free society, and its drives push in a wholly different direction.

THE WORLD IS NOT TOO COMPLICATED

We have now explored AI as a continuation of many previous technological, economic, and political tendencies. Now we've arrived at a point of payoff, so to speak. Implicit in our perspective were kernels for explaining a number of other phenomena that we've been thinking about throughout this book: why did the internal functioning of our tools increasingly vanish from the social view? Why have bureaucratic-managerial structures ballooned wherever we look? Why is the working class so stratified? Why is it that the more humanity's technological and scientific powers increase, the less we feel like we can actually control those powers? And perhaps we can even shed some new light on why it is so hard to imagine an alternative to capitalism.

The story of the transistor illustrated the changing nature of our tools: technological developments have increasingly severed our relationship to our tools' internal functioning. We suggested that we experience this as an increase in complexity. But what is that increasing and mysterious "complexity" other than knowledge that is purposefully embedded in the machine or the managerial apparatuses, in order to make it unnecessary for workers and users, for us, to understand it. After all, for every complex aspect of our technology, there is *someone* who understands, who does know. Complexity is decidedly not the same as genuine incomprehensibility. That is not to say that the scale and differentiation of new scientific and technological knowledge have not played a part. But under capitalism, new knowledge, and in particular knowledge of tools and labor processes is continually and systematically put on a need-to-know basis, and often sealed off, as an effect of deskilling.

Knowledge gets concentrated among managers or technobureaucrats, so as to deskill, to cheapen the labor of others. The unequal distribution of knowledge in society is an effect of deskilling—and, insofar as that unequally distributed knowledge is also continually used in new technological and scientific developments, it also produces the affordances for new rounds of deskilling. In other words, under the pressures of capitalism, technological developments and the sealing off of knowledge are in a self-reinforcing feedback loop. And since all that knowledge is systematically put

on a need-to-know basis, monopolized by some, it cannot appear as anything other than complexity to everyone else—as knowledge possessed only by someone (or even something) else.

The vanishing of our tools' comprehensibility is not simply an accidental byproduct of technological development. The closing off of our tools from view serves a particular purpose—to increase the power of one class, the owners of those tools, over the other one, those who work with them. These economic pressures and developments end up affecting all kinds of tools, whether they are capital or consumer goods. From computers to DNA testing services, many things do, after all, transition from one domain to the other, as their prices fall, and new uses are invented. If the backs of our smartphones are literally and figuratively glued shut, then this is not simply an inevitable side effect of technological development, but about locking knowledge behind closed doors, and sealed backs, about putting knowledge on a need-to-know basis. In other words, it is the fact that knowledge is power—the power to bargain for higher wages, and better working conditions, or the power to decrease wages, and increase capital's profit and control over work—that made our tools into what they are: objects whose complexity enables us to be ever more productive, but that also continually shift power from workers to capital.

The vanishing of our tool's internal functioning went hand in hand with a rise of innumerable and ever larger bureaucracies, even as we lived in times dominated by a neoliberal self-justification of capitalism that perpetually claims to be opposed to bureaucracies. Such justifications even claim that capitalism and bureaucracies are like oil and water, almost akin to logical opposites. Now, we know that the freest of capitalist markets isn't inimical to bureaucracies that hoard knowledge in order to create control. Quite the contrary—capitalism incessantly and by necessity produces such bureaucracies. Competition drives corporations to continually develop new technologies, in order to lower production costs per piece. This can be achieved not only through increased productivity, but also through deskilling—to the capitalist, they are simply cost reductions, reductions in input variables, indistinguishable from the only thing that ultimately matters: prices and profits. And thus, wherever new technology makes increases

in productivity economically feasible, that's what it will do—and wherever deskilling is economically feasible, *that's* what it will do.

Deskilling, however, requires that knowledge be put on a need-to-know basis, transferred away from workers and embedded in managerial and specialist bureaucracies. At first, this may only create managerial bureaucracies. But soon, even engineers that started out as purely technical specialists are more and more involved in the planning of production processes, in gathering knowledge to exercise control. Wherever oversight is transformed into a technological property of tools (from the steam engine to the Amazon AI), engineers become techno-managers. Nowhere are engineering departments concerned solely with productivity, they are concerned with costs per piece, and hence they turn into techno-bureaucracies. Management and technical specialists start forming mixed bureaucracies: technology and management, tools and control are ever more deeply intertwined. And whenever those bureaucratic structures get big enough, the same principles of deskilling are applied internally as well. A process of recursive deskilling and bureaucracy creation unfolds, and is unfolding every day. The managerial pyramid turns into a professional-managerial pyramid. It grows incessantly, as long as it can keep overall wages down and make sure that the ever increasing productivity is turned into profit—and never into social wealth, into more interesting and engaging workplaces, or into simply more free time.

We began with a very abstract description of what class is: to be a member of the working class is to live off wages, to sell one's ability to work, so that someone else can put it to use for their profit. To be a member of the capitalist class is to live off the profits generated by capital, by assets (often the kind of assets that require someone else's work), rather than by selling one's ability to work. But of course, the real, lived experience of class is much more varied than that, the class of people who depend on wages is not internally uniform, but extremely stratified. In fact, among people who work for a wage in the US, the majority would likely consider themselves "middle class," and not "working class." We can now connect those two kinds of descriptions—stratification is inherent to the professional-managerial pyramid, which concentrates knowledge and control over work processes,

at every level. It's the recursive nature of the drive to deskill that gives rise to stratification, as it applies wherever profits and control can be increased by deskilling many and upskilling a few. In divvying up labor processes, capitalism increasingly divvies up the working class itself.

The more divided the working class is, the harder it is for them to find a common interest and common ground. This is obvious at the level of wages (where division undermines collective bargaining power, by separating out different interests). But it also applies at the level of politics, which is dominated by shifting alliances that sections of capital make with strata of the working class. Of course, we have only touched upon many of these developments, and viewing the working class as stratified by their access to economically valuable knowledge has many potentially useful applications that we won't be able to get to in this book. Consider, for instance, the right-wing attempts to channel anger among some working-class people away from collective conflict with their bosses, and into anti-intellectualism. There are ongoing attempts from the right to declare a barista with a college degree and \$100k of student debt the "elite," while the inheritor of a used car dealership, a small capitalist that makes seven figures in profit per year, is declared "working class."

Earlier in this book, we had asked why humanity, having gained the awesome power to change the climate of our whole planet, simultaneously seems to have lost the power *not* to change it. Or more broadly, why is it that the more scientific and technological power we gain, the less able we seem to control those powers? In that chapter, we talked about the way that capitalism imbues our tools, as capital, with a kind of motive force of their own, the drive for profit. There, we said that this drive for profit is a structural pressure, an organizing principle of our society—it could not exist without a capitalist class, without managerial executors, but capital's drive to profit does not depend on any particular person's desires (it even shapes the desires of people to match its own requirements). Or in other words: capitalism is a system that dominates everyone inside of it, even the capitalist class is subject to its pressures. And when tools are capital, they are things that exert that pressure, the pressure to use them to

make a profit, wherever and whenever possible—even if it means trashing the climate.

We can now add a second layer to that account: from the perspective of the working class, tools really do represent not the power to do something, but the power that others (the capitalist class, and their professional-managerial representatives) have over them. We quoted Harry Braverman's paradox earlier: "The more science is incorporated into the labor process, the less the worker understands of the process." This is the mystery solved—why does an increase in our scientific understanding limit, rather than enhance, our ability to act on the world in accordance with our will? Why have we gained awesome powers, but lost the power to not use them? Because the science and the tools that grant humanity these powers, are simultaneously weapons wielded by one class of people, the class of capitalists, in order to control everyone else in the name of profit. The manifold applications of science, the tools that capitalism continually develops are also Taylor(ism)-made to control and cheapen people's labor. Science is used to hide, privatize, and monopolize science. The more knowledge we create, the more effective the capitalist class becomes at putting much of it on a need-to-know basis. The more sophisticated the tool gets, the more bureaucracies pop up to control it for capital, and the more its internal workings are hidden from the social view. The more science is embedded in our tools and labor processes, the less we can control them. They, instead, control us, so that capital can make its most profitable use of the one thing that it buys from us—our ability to work. In other words, the 'humanity' that has gained the power to change the planet and has lost the power not to, that's the capitalist class. The capitalist class has gained power, but it really is power that they cannot not use (at least not if they want to remain capitalists). It's a power wielded against the 'nonhumanity,' the working class, the rest of us—we have never lost it, we have simply not yet taken it away from those who want to use it to make a profit, no matter what.

Finally, why did so many of us end up believing Thatcher's dictum that "there is no alternative"? Where did the old utopias of a deeply democratic organization of society and of work disappear to? Why do markets, wage labor, and profits seem like

the only way to organize society, even though it makes so many lives miserable, and is threatening to make our planet uninhabitable for humans? Why can we no longer imagine that we could, simply put, make do without bosses, without managers, without capitalists? Why could neoliberalism—in the words of Naomi Klein—“convince us that we are not just incapable of self-preservation but fundamentally not worth saving”?²⁰ Why could it convince us that only the market, only capital can know how to organize the world? Because capital has indeed monopolized the knowledge of how the world is organized today, concealing knowledge wherever it can, revealing only what it must. By embedding knowledge in tools, capitalism privatizes not just the instruments that we use in making the many things that we need to live, from food, to housing, to art, to care. If the tools’ internal structure is hidden from view, privatized, its workings put on a need-to-know basis, how can we, the class that makes all the tools, imagine ourselves as making do without capital? As the internals of our tools vanished, so did the fact that all the things we consume have been produced by us, by the working class. We live in a world where goods seem to simply appear in the store. Many of them do so in ways that make it not only unnecessary to understand how they were made, how they got there, but difficult, or even impossible. This, too, is an effect of deskilling, of privatized knowledge—the sense that objects no longer have a history of their production *by us*.

And yet, as we understand these issues, as we unveil the relation between economics and power, between tools and politics, we can also discover new possible paths. As workers, we must struggle collectively not only for wages, but for our ability to directly control and comprehend the labor processes that we are engaged in. Those are the sources from which we can develop our collective strength, the strength we need even in the struggle over wages. They are also the locus at which economic struggles become clear as political struggles—not just struggles over how we distribute the pie, but how we decide on which pies to make, and whether the world should really be an oven.

Wherever we form or join unions, we should strive to elevate the collective control over knowledge, the democratization of it in

the workplace as one of our main goals. It is up to all of us to take on this task, to develop better ways of sharing, distributing knowledge, of de-privatizing it. We need to make knowledge into tools of our collective agency, rather than private tools for deskilling, stratification, and capitalist control. We need democratic control over and collective understanding of our own labor processes.

The fact that the world appears as “too complicated” is a real veil that is at the heart of the disappearance of classical utopias. But we do not need to stand behind it. Capital’s power to monopolize knowledge is not based on some mysterious power that only it can wield, it comes from organization and is based in bureaucracies. If society decided to (if the working class decided to), we could abolish corporations, take them over, and make them into institutions that are cooperatively run by the workers, run so as to free knowledge, not to hide it, to give people knowledge and control over their own work, rather than centralizing knowledge and control in capital. This is a question of imagination, and of collective will—a question for the political animal, not for the toolmaker. We can change the world, but we can only do so together. We need to organize ourselves, and gather the knowledge with which we can build a better world.